

Warm-up

(1) Compute $\int_1^\infty \frac{1}{x^4} dx$.

(2) Compute $\int_0^4 \frac{1}{\sqrt{x}} dx$.

Medium

(1) Compute $\int_1^2 \frac{dx}{x \ln(x)}$.

(2) Compute $\int_{-\infty}^1 \frac{3}{x^2 + 1} dx$.

(3) Compute $\int_0^\infty \frac{1}{x^2} dx$.

(4) Compute $\int_0^{\pi/2} \frac{\sin(x)}{\sqrt{\cos(x)}} dx$.

(5) Compute $\int_1^\infty \frac{dx}{\sqrt{x} + x\sqrt{x}}$.

Stretch

(1) Define the *gamma function* by

$$\Gamma(x) = \int_0^\infty t^{x-1} e^{-t} dt.$$

Use the comparison test to show that this converges for $x > 0$. Depending on the value of x , you may need to do two comparison tests.

(2) Compute $\Gamma(1)$, and then integrate by parts to derive the identity $\Gamma(x+1) = x\Gamma(x)$.